

Chapter 9 — Edge computing and the IoT pipeline

Raw sensor streams overwhelm networks if everything ships to the cloud

Learning objectives

- By the end of this part, learners should be able to outline stages of an IoT data pipeline

Pipeline stages from generation to analytics

- An IoT pipeline typically moves through data generation at the sensor
- Each stage must be designed so volume and delay do not erase the value of continuous

Example 9.17 — Smart Thermostat Local Reading

- Example 9.17 — hands-on module
- Example 9.17 gives a minimal sensing vignette
- That local reading is the generation step before gateway or cloud processing
- Explore the chapter example module
- View files: ``modules/chapter9/example17/``

Edge processing in the pipeline

- Edge computing processes data at or near the device
- Doing so reduces load on central servers and shortens delay for real-time decisions
- Transmission may still use Wi-Fi

Benefits: latency, bandwidth, reliability

- Processing closer to the source cuts latency for time-sensitive applications
- Sending only relevant or pre-processed data optimizes bandwidth and cost
- Edge devices can also keep working when connectivity drops
- Together these benefits make continuous IoT collection operationally viable

Example 9.18 — Edge Processing for Traffic Lights

- Example 9.18 — hands-on module
- Example 9.18 shows smart traffic systems that process camera and sensor data at the edge
- Local decisions keep control loops tight
- Explore the chapter example module
- View files: ``modules/chapter9/example18/``

Example 9.19 — Real-Time Logistics Rerouting

- Example 9.19 — hands-on module
- Example 9.19 applies IoT tracking to logistics
- Timely sensing plus responsive processing turns location streams into operational
- Explore the chapter example module
- View files: ``modules/chapter9/example19/``

Takeaways

- IoT pipelines span generation, transmission, edge and cloud processing, and analytics
- Edge computing improves latency, bandwidth use, and offline resilience
- Thermostats, traffic lights

Next

- Complete the quiz for this part
- The next part covers scalability, security